NATIONAL UNIVERSITY



Third Year Syllabus

Department of Statistics

Four Year B.Sc Honours Course
Effective from the Session : 2013–2014

National University

Subject: Statistics
Syllabus for Four Year B. Sc Honours Course
Effective from the Session: 2013-2014

Year wise Papers and marks distribution

THIRD YEAR

Paper Code	Paper Title	Marks	Credits
233601	Sampling Distribution	100	4
233603	Statistical Inference	100	4
233605	Regression Analysis	100	4
233607	Design of Experiments	100	4
233609	Applied Statistics	100	4
233611	Introduction to computer and SPSS	100	4
233612	Lab 5: Statistical Inference	50	2
233614	Lab 6: Regression Analysis	50	2
233616	Lab 7: Design of Experiments	50	2
233618	Lab 8: Applied Statistics	50	2
	Total =	800	32

Detailed Syllabus

Paper Code	233601	Marks:100	Credits: 4	Class Hours: 60 hrs.
Paper Title:	Sampling Distributions			

Functions of Random Variables and Random Vectors: Methods of distribution functions, Method of transformations. Method of moment generating functions. Probability integral transformation. Distribution of sum, difference, product and quotient of random variables, Functions of random vectors of continuous and discrete type. Sampling from normal distributions. Role of normal distribution in statistics, Distribution of sample mean, sample variance and their independence, Detailed study of $\chi 2$, t and F distribution, Standard errors of statistics and their large sample approximations, Transformation of variates. Non-central $\chi 2$, t and F-distributions.

Order Statistics: Definition and distributions of functions of order statistics for discrete and continuous case, asymptotic distributions, sample cumulative distribution functions. Joint distribution of *n* order statistics. Joint distribution of two order statistics, Distribution of range and some other statistics.

Books Recommended:

1 Ashraf Ali :Theory of Statistics Vol-2
2 Rohatgi :Statistical Inference
3 Steel and Torre :Principle and Procedures of Statistics
4 Kapur, J.N. and Saxena, H.C :Mathematical Statistics

Paper Code	233603	Marks:100	Credits: 4	Class Hours: 60 hrs.
Paper Title:	Statistical Inference			

Part A : Estimation

Concept of statistical inference. Concept of estimation and problem of estimation, Estimator and Estimate. Properties of a good estimator: Unbiasedness, Consistency, Efficiency, Sufficiency, Minimum Variance, Estimator, Pitman-closer. Pitman-closest.Invariance, Mean Square Error. LMVUE.. UMVUE. Rao-Black Well theorem, Cramer-Rao Inequality, MVB estimator, Use of Cramer-Rao inequality in finding UMVUE.

Point Estimation: Method of Maximum likelihood. Method of moments. Method of Least Square. Method of minimum chi square. Bayesian method of estimation.

Interval Estimation: Confidence interval and confidence limit. Confidence belt and Confidence bound. Confidence coefficient, Methods for finding confidence interval, confidence interval for mean, variance, difference of mean, ratio of variance, correlation coefficient, regression coefficient.

Part B: Test of Hypothesis

Concepts and Fundamentals: Problem of testing hypothesis, logic behind testing of hypothesis. Hypothesis; simple and composite, null and alternative. Test function, level of significance, p-value, test errors, size of errors, power of a test, one-tailed and two-tailed test. Concept of test Statistic. Normal, $\chi 2$, *t* and F statistics. Testing the significance of a single mean, single variance, single proportion, single correlation coefficient Difference of two means and proportions, two correlation coefficients. Ratio of two variances and their confidence intervals. Paired t-test. Testing the homogeneity of several population

means, variance and proportions, Correlation coefficients. Test of goodness of fit. Association of Attributes : Concepts of independence, association and disassociation. Derivation of frequency $\chi 2$, uses and limitations: Best critical region, most powerful test, unbiased test, Neyman-pearson lemma. Uniformly most powerful test. Definition, Properties, uses, mean and variance test.

Non parametric Test: Run test, sign test (one sample, two sample) Median, Rank test, Mann Whitny U-test.

Books Recommended:

1 Hogg, R.Y. and Craig, A. T: Introduction to Mathematical Statistics
2 Fraser, D.A.S. Non-Parametric, P.U. -Statistical Inference
3 Islam, M. N. An Introduction to Statistics and Probability, 4th Ed, Mullick & Brothers
4 Sexena, H.C. and Surendran, P. U Statistical Inference
5 Kendal and Stuart The Advanced Theory of Statistics
6 Snedecor Statistical Methods
7 K. C Bhuyan Probability Distribution Theory and statistical Inference

Paper Code	233605	**Marks:100**	**Credits: 4**	**Class Hours: 60 hrs.**
Paper Title:	**Regression Analysis**			

Introduction: Historical background, definition and uses. Formulation of multiple linear regression model with full rank, assumptions, Estimation of parameters & their properties. Coefficient of determination, Test of significance and confidence intervals of the parameters. Multiple regression models, Regression model in matrix notation, Model assumptions. OLS estimation of regression parameters. Large and small sample properties of estimators. Properties of residuals. Constrained Least squares, Generalized & Weighted least squares, Partitioning of total sum of squares into components, Coefficient of multiple determination (R2, Adjusted R2, Properties and sampling distributions of component *SS.* Testing general regression model, Confidence intervals for individual parameter, Lack of fit. Polynomial regression & Orthogonal polynomials.
Detail study of multiple and partial correlation
More on Checking Fitted Models: Detecting outliers using ordinary residuals, the hat matrix H and the various types of residuals, detection of influential observations: Cook's statistics, other statistics measuring influence.

Selecting the "Best" Regression Equation: All possible regression and "best subject" regression, backward elimination, significance levels for selection procedures.

Robust Regression : Motivation of Robust regression, Breakdown point and Robust estimators, Computation of the least median of squares and the reweighted Least squares residuals for simple regression. Properties of the LMS, the S-estimators. Outlier diagnostics for single and multiple cases.

Indicator/ Dummy Variables: Indicator variables versus regression on allocated codes, dummy variables to separate blocks of data with different intercepts same model interaction terms involving dummy variables Dummy variables for segmented models.

Books Recommended:

1 Draper, N. R. and Smith, H. Applied Linear Regression

2	Montgomery, D. C. and Peck, E.	An Introduction to Regression Analysis
3	Seber, G.A.F	Linear Regression Analysis
4	Ashraf Ali	Theory of Statistics Vol-2
5	Rohatgi	Statistical Inference
6	Steel and Torre	Principle and Procedures of Statistics
7	Kapur, J.N. and Saxena, H.C	Mathematical Statistics

Paper Code	233607	**Marks:100**	**Credits: 4**	**Class Hours: 60 hrs.**
Paper Title:	**Design of Experiments**			

Introduction: Concept of design, design of experiments and different terms, Different types of models. Analysis of variance of one-way, two way and three-way classified data with single observation per cell. Construction and analysis of basic designs: CRD, RBD and LSD, RBD with multiple but equal observation per cell. Replicated LSD. Estimation of missing value and analysis. Relative efficiency, Multiple comparisons.

Factorial Experiments: Factorial experiments of the type 2n and p×q and their analysis of variance. Yates algorithm. Confounding total and partial. Their construction and analysis of variance. Orthogonal latin square design. Gracco LSD. Split plot design, Strip-plot design.

Book Recommended:

1	Montgomery. D.C	Design and Analysis of Experiment
2	Das M.N & Giri N.C	Design and Analysis of Experiment
3	Yales, F	Design and Analysis of Factorial Experiment
4	Cox, D.R	Planning of Experiment
5	Scheffc. H	Analysis of Variance
6	Mallick. SA	পরিসংখ্যানের ডিজাইন
7	Ship RN & Debnatts SC	An introduction of Theory of Statistics
8	K. C Bhuiyan	পরীক্ষনে নকশা (১ম ও ২য খন্ড)

Paper Code	233609	**Marks:100**	**Credits: 4**	**Class Hours: 60 hrs.**
Paper Title:	**Applied Statistics**			

Economic Statistics: Determination of Demand and Supply curve from market data. Demand function. Elasticity of Demand. Utility function. Maximization of utility. The choice of utility Index, Indifference curves. Rate of commodity substitution. Income effect. Substitution effect. Return to scale. Marginal propensity to consume. Basic concepts of the theorem of Economic growth. .Distribution of Income. Pareto income distribution, Lognormal distribution, Lorenz curve, Gini's concentration ratio, Engle's law, National Income: Measurement and problems.

The Production Function: Productivity curves, Isoquants. Shape of the production function, Elasticity of substitution. Optimization behaviors. Cobb Doglus and C.E.S. production function.

Index Number: Definition, problem, Construction of simple and weighted indices, cost of living index number, Test of index number, uses of index number.

Time Series Analysis: Different components of time series data, Measurement of secular trend and seasonal variation. Spencer's I5-points and 2I-point formulae. Correlograms analysis. Evolutatory and stationary time series. Different schemes which account for oscillation in a stationary series and their detailed description.

SQC: Quality control measures, statistical quality control techniques, its uses and usefulness. Product control and process control. Assignable and non assignable causes of variations. Basic control charts mean, standard deviation, range no. of defective, no. of defects. Acceptance sampling, Derivation of sampling plans (single and double) OC, AOQ and ASN curves.

Book Recommended:

1	Singh Parashar and Singh	Econometrics and Mathematical Economics
2	Davis	Economic Statistics
3	Henderson and Quandit	Microeconomic Theory
4	Box and Jankins	Time Series Analysis
5	Chatfield, C.	Analysis of Time Series, An Introduction
6	Grant, EL	Statistical Quality Control

Paper Code	233611	**Marks:100**	**Credits: 4**	**Class Hours: 60 hrs.**
Paper Title:	**Introduction to Computer and SPSS**			

Importance and application of computer, Classification of computers, computer organization, function of central processing unit (CPU), Input & Output devices, memory organization-primary storage, secondary storage.

RAM and ROM, Hardware and software, system software and application software, hard disk and floppy disk, magnetic disk and magnetic tape. OCR and OMR, batch processing, Buffer, microprocessor, network, internet, Programming languages, machine language, assembles language, microinstructions, high- level programming languages, assembler and compiler, interpreter.Basic idea of operating system, preliminaries of DOS and WINDOWS.

Overview of SPSSWIN, Operation commands, data definition and manipulation commands, file management, commonly used procedure commands for data analysis Computing descriptive statistics, correlation, regression, comparing group means, analysis of categorical data, statistical tests, graphical representation, data plotting using SPSSWIN.

Book Recommended:

1	Zikmaund, William, G	Business Research Methods. 6th Edition
2	Cooper, Donald, R	Business Research Methods. 5th Edition.
3	Scheid, F	Introduction to Computer Science

4 Walker, Terry, M Introduction to Computer Science
5 Duncan, R- Advanced MS DOS Programming Microsoft Press
6 Sander Dorold, H Computer today.

Paper Code	233612	**Marks:50**	**Credits: 2**	**Class Hours: 30 hrs.**
Paper Title:	**Lab-5: Statistical Inference**			

Fitting of curves (t, F & $\chi 2$). Test of large samples other common tests (single means, variances correlations coefficients and several means, variances and correlation co-efficient). Paired t-tests, Construction of confidence intervals. Non-parametric test (sign test, run test, median test, mann-whitney test etc).

Problem on Maximum likelihood, least squares, method of moments for various distributions.

Paper Code	233614	**Marks:50**	**Credits: 2**	**Class Hours: 30 hrs.**
Paper Title:	**Lab-6: Regression Analysis**			

Estimation of parameters of simple and multiple regression model. Test for parameters. Inference of multiple regression model. (Test of significance of the overall regression). Partial correlation, multiple correlation. Fitting of polynomial regression. Orthogonal polynomial. Selecting the best regression model.

Paper Code	233616	**Marks:50**	**Credits: 2**	**Class Hours: 30 hrs.**
Paper Title:	**Lab-7: Design of Experiment**			

Analysis of variance: (One way and two way classification with single observation per-cell) CRD, RBD, LSD, Missing plots estimation and analysis. Relative efficiency of different designs. Factorial experiment 22, 23, 24, and confounding. p×q factorial experiments. Split plot design

Paper Code	233618	**Marks:50**	**Credits: 2**	**Class Hours: 30 hrs.**
Paper Title:	**Lab-8: Applied Statistics**			

Fitting of Parreto, Lognomal, Curve, Determination of concentration ratio. Determination of trend values and seasonal indices by different methods. Computation of index numbers and cost of living index numbers. Control charts (variable & attribute). Derivation of sampling plan (single and double). OC curve and ASN curves.